1 I

1.1 (1)

The electric and magnetic fields of the plane electromagnetic wave are given by:

$$\vec{E}(\vec{x},t) = \vec{E}_0 \exp[i(\vec{k}\cdot\vec{x} - \omega t)],$$
$$\vec{B}(\vec{x},t) = \vec{B}_0 \exp[i(\vec{k}\cdot\vec{x} - \omega t)].$$

When applying differential operators to these fields, we obtain:

$$\nabla\cdot\vec{E} = i\vec{k}\cdot\vec{E},$$
$$\nabla\times\vec{E} = i\vec{k}\times\vec{E},$$
$$\nabla^2\vec{E} = -k^2\vec{E},$$
$$\frac{\partial\vec{E}}{\partial t} = -i\omega\vec{E}.$$

The same relations hold for $\vec{B}$, with $\vec{E}$ replaced by $\vec{B}$.

Thus, using $\nabla\cdot\vec{E} = 0$ and $\nabla\cdot\vec{B} = 0$, we obtain:

$$\vec{k}\cdot\vec{E}_0 = 0, \qquad \vec{k}\cdot\vec{B}_0 = 0.$$

1.2 (2)

Using Faraday's law and Ampère's law:

$$\nabla\times\vec{E} = -\frac{\partial\vec{B}}{\partial t}, \qquad \nabla\times\vec{B} = \epsilon_0\mu_0\frac{\partial\vec{E}}{\partial t},$$

we can respectively obtain:

$$\vec{k}\times\vec{E}_0 = \omega\vec{B}_0,$$
$$\vec{k}\times\vec{B}_0 = -\epsilon_0\mu_0\omega\vec{E}_0.$$

1.3 (3)

Taking the curl of both sides of Faraday's law, we obtain:

$$\nabla\times(\nabla\times\vec{E}) = -\frac{\partial}{\partial t}(\nabla\times\vec{B}) = -\epsilon_0\mu_0\frac{\partial^2\vec{E}}{\partial t^2}.$$

Using the vector identity

$$\nabla\times(\nabla\times\vec{E}) = \nabla(\nabla\cdot\vec{E}) - \nabla^2\vec{E} = -\nabla^2\vec{E},$$

we finally get:

$$\left(\nabla^2 - \epsilon_0\mu_0\frac{\partial^2}{\partial t^2}\right)\vec{E} = 0.$$

Similarly, we can obtain:

$$\left(\nabla^2 - \epsilon_0\mu_0\frac{\partial^2}{\partial t^2}\right)\vec{B} = 0.$$

Comparing these equations with the standard wave equation:

$$\left(\nabla^2 - \frac{1}{v^2}\frac{\partial^2}{\partial t^2}\right)\vec{u} = 0,$$

we find that the propagation speed of both the electric and magnetic fields is:

$$v = \frac{1}{\sqrt{\epsilon_0 \mu_0}}.$$

2 II

2.1 (4)

Following the same procedure as in (3), we obtain:

$$\begin{aligned}\nabla \times (\nabla \times \vec{E}) &= -\frac{\partial}{\partial t}(\nabla \times \vec{B}) \\ &= -\epsilon\mu\frac{\partial^2 \vec{E}}{\partial t^2} - \mu\frac{\partial \vec{j}}{\partial t}.\end{aligned}$$

Furthermore, using Ohm's law ($\vec{j} = \sigma\vec{E}$) and $\nabla \cdot \vec{E} = 0$, we obtain:

$$\left(\nabla^2 - \epsilon\mu\frac{\partial^2}{\partial t^2} - \sigma\mu\frac{\partial}{\partial t}\right)\vec{E} = 0.$$

2.2 (5)

Substituting the electric field

$$\vec{E}(\vec{x}, t) = \vec{E}_0 \exp[i(kz - \omega t)]$$

into the wave equation obtained in (4), we get:

$$k^2 = \epsilon\mu\omega^2 + i\mu\sigma\omega.$$

2.3 (6)

Under the quasi-static current approximation $\epsilon\omega/\sigma \ll 1$, the dispersion relation reduces to:

$$k^2 = i\sigma\mu\omega.$$

Let $k = k_1 + ik_2$, where k_1 and k_2 are real:

$$(k_1^2 - k_2^2) + i(2k_1k_2) = i\sigma\mu\omega.$$

Considering the propagation direction and the attenuation of the wave inside the conductor, we have $k_1 > 0$ and $k_2 > 0$. Hence:

$$k_1 = k_2 = \sqrt{\frac{\sigma\mu\omega}{2}}.$$

2.4 (7)

Substituting

$$k = \sqrt{\frac{\sigma\mu\omega}{2}}(1+i)$$

into the expression for $\vec{E}$, we obtain its amplitude:

$$E(z) = E_0 \exp\left(-\sqrt{\frac{\sigma\mu\omega}{2}}z\right).$$

2.5 (8)

The distance d at which the amplitude becomes $1/e$ of its initial value satisfies:

$$\boxed{d = \sqrt{\frac{2}{\sigma\mu\omega}}}$$

Using the given parameters, we obtain:

$$\begin{aligned}\sigma &= \frac{1}{3.6\times10^{-8}\,\Omega\cdot\mathrm{m}},\\ \mu &= 1.2\times10^{-4}\,\mathrm{H\,m^{-1}},\\ \omega &= 2\pi\times955\times10^{3}\,\mathrm{s^{-1}}.\end{aligned}$$

Therefore:

$$d \simeq 1\times10^{-5}\,\mathrm{m}.$$

3 III

3.1 (9)

Taking the curl of both sides of

$$\frac{\partial\vec{j}}{\partial t} = \frac{1}{\Lambda}\vec{E},$$

and using Faraday's law, we obtain:

$$\frac{\partial}{\partial t}(\nabla\times\vec{j}) = -\frac{1}{\Lambda}\frac{\partial\vec{B}}{\partial t}.$$

Thus:

$$\boxed{\frac{\partial}{\partial t}\left(\nabla\times\vec{j}+\frac{1}{\Lambda}\vec{B}\right)=0}$$

3.2 (10)

For static electromagnetic fields, Ampère's law reduces to:

$$\nabla\times\vec{B} = \mu\vec{j}.$$

Therefore:

$$\vec{j} = \frac{1}{\mu}\nabla\times\vec{B}.$$

Substituting this into the London equation:

$$\nabla \times \vec{j} + \frac{1}{\Lambda}\vec{B} = 0,$$

we obtain:

$$\frac{1}{\mu}\nabla \times (\nabla \times \vec{B}) + \frac{1}{\Lambda}\vec{B} = 0.$$

Using the vector identity and $\nabla \cdot \vec{B} = 0$:

$$\nabla \times (\nabla \times \vec{B}) = -\nabla^2 \vec{B},$$

we finally get:

$$\nabla^2 \vec{B} = \frac{\mu}{\Lambda}\vec{B}.$$

Since the interface is the xy-plane and the applied magnetic field is uniform and parallel to the interface, the system is translationally invariant along the x and y directions. Therefore, the magnetic field is independent of x and y. Moreover, by symmetry, it has only an x-component. Thus, we can write:

$$\vec{B}(\vec{x}) = (B_x(z), 0, 0).$$

Here, B_x denotes the x-component of the magnetic field, while z indicates that this component varies with the distance from the interface.

Consequently:

$$\nabla^2 \vec{B} = \frac{d^2 B_x(z)}{dz^2}\hat{x}.$$

Substituting this into the previous equation, we obtain:

$$\frac{d^2 B_x(z)}{dz^2} = \frac{\mu}{\Lambda} B_x(z).$$

The general solution is:

$$B_x(z) = C_1 e^{\sqrt{\mu/\Lambda}z} + C_2 e^{-\sqrt{\mu/\Lambda}z}.$$

Since the magnetic field must remain finite as $z \to \infty$, we have $C_1 = 0$. Applying the boundary condition $B_x(0) = B_0$, we obtain:

$$\boxed{B_x(z) = B_0 e^{-\sqrt{\mu/\Lambda}z}}$$